

Practical Quantum Subspace Expansion: Stability, Noise, and Scaling

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Abstract

Keywords

quantum subspace expansion; variational quantum eigensolver; excited states; active spaces; measurement noise

Abbreviations

QSE, quantum subspace expansion; VQE, variational quantum eigensolver; UCC, unitary coupled cluster; CASCI, complete active space configuration interaction.

Introduction

The Quantum Computing Edge in Quantum Chemistry

1. Discuss why quantum computing has an edge in quantum chemistry. Relate to the exponential scaling that classical computers will face. 2. Talk about how quantum computers will also not be perfect, mainly limited by statistical noise and hardware errors (lack of error correcting code). + scaling also can be rough even if its not as bad as classical 3. Discuss the various advances on generating excited states (briefly explain why we are interested in it). Mainly need to answer the question, WHY subspace expansion? Why QSE?

Quantum Subspace Expansion for Excited States

Add a diagram for workflow?

Quantum Subspace Expansion is a subspace exploration method, that builds a subspace around a guess ansatz. This is typically done by first obtaining a ground state reference state via a Variational Quantum Eigensolver (VQE). This is another hybrid-quantum method that uses a classical optimiser to optimise coefficients of circuit parameters.

In this study, the Variational Quantum Eigensolver (VQE) algorithm was employed to approximate a ground reference state ψ_V . 6 different classes of Unitary Coupled Cluster (UCC) Ansatz were applied, namely UCCSD, UCCSDSinglet, UCCGSD and kUpCCGSDSinglet ($k = 1, 2, 3$). Each Ansatz is characterised by a unitary Operator defined as

$$\hat{U}_{\text{UCC}}(\boldsymbol{\theta}) = \exp\left(\hat{T}(\boldsymbol{\theta}) - \hat{T}^\dagger(\boldsymbol{\theta})\right) = \exp\left[\sum_{\lambda} \theta_{\lambda} \left(\hat{T}_{\lambda} - \hat{T}_{\lambda}^\dagger\right)\right]. \quad (1)$$

whereby $\hat{T}$ is a parameterised excitation operator. Under the exponent, its self adjoint $\hat{T}^\dagger$ is subtracted, rendering the UCC operator unitary. The UCC operator was applied to the single reference Hartree Fock (HF) state to get the ground reference state.

$$|\Psi_{\text{UCC}}(\boldsymbol{\theta})\rangle = \hat{U}_{\text{UCC}}(\boldsymbol{\theta}) |\Phi_{\text{HF}}\rangle. \quad (2)$$

Given an optimized reference state $|\Psi_0\rangle$, QSE constructs a non-orthogonal projected basis by applying a set of expansion operators $\{\hat{O}_i\}$,

$$|\Phi_i\rangle = \hat{O}_i |\Psi_0\rangle. \quad (3)$$

The Hamiltonian and overlap matrices in this projected basis are

$$H_{ij} = \langle \Phi_i | \hat{H} | \Phi_j \rangle = \langle \Psi_0 | \hat{O}_i^\dagger \hat{H} \hat{O}_j | \Psi_0 \rangle, \quad (4)$$

$$S_{ij} = \langle \Phi_i | \Phi_j \rangle = \langle \Psi_0 | \hat{O}_i^\dagger \hat{O}_j | \Psi_0 \rangle. \quad (5)$$

The QSE energies are obtained by solving the generalized eigenvalue problem

$$\mathbf{H}\mathbf{c} = E\mathbf{S}\mathbf{c}. \quad (6)$$

In practice, the overlap matrix is first diagonalized as

$$\mathbf{S} = \mathbf{V}\mathbf{\Lambda}\mathbf{V}^\dagger, \quad \mathbf{V}^\dagger\mathbf{V} = \mathbf{I}. \quad (7)$$

This defines the orthogonalizing transformation

$$\mathbf{X} = \mathbf{V}\mathbf{\Lambda}^{-1/2}, \quad \mathbf{X}^\dagger\mathbf{S}\mathbf{X} = \mathbf{I}. \quad (8)$$

Using $\mathbf{c} = \mathbf{X}\mathbf{y}$, the generalized eigenvalue problem is converted into an ordinary Hermitian eigenvalue problem,

$$\mathbf{X}^\dagger\mathbf{H}\mathbf{X}\mathbf{y} = E\mathbf{y}, \quad \mathbf{c} = \mathbf{X}\mathbf{y}. \quad (9)$$

The eigenvector matrix is first multiplied by the whitening factor $1/\sqrt{\lambda_i}$. Crucially, this means that when λ_i is small, this factor becomes large. As such, noise and errors in the sampled matrix elements can be amplified greatly. This is the heart of what many consider the instability problem that plagues the generalized eigenvalue problem. This problem cannot be easily ignored because statistical sampling noise is inherent to quantum computers.

Therefore, in practice, thresholding is commonly used. Tiny overlap eigenvalues below a threshold τ are often dropped to prevent numerical instabilities. If $\mathbf{V}_r$ and $\mathbf{\Lambda}_r$ contain only the retained eigenvectors and eigenvalues with $\lambda_i > \tau$, the solved problem becomes

$$\mathbf{X}_r^\dagger\mathbf{H}\mathbf{X}_r\mathbf{y} = E\mathbf{y}, \quad \mathbf{X}_r = \mathbf{V}_r\mathbf{\Lambda}_r^{-1/2}. \quad (10)$$

This reduces the number of solutions that can be recovered and lowers the effective rank of the QSE subspace.

Current QSE Studies and Contributions of This Work

Results and Discussion

Effective Dimensions of the QSE Subspace

Subspace Completeness Beyond Ansatz Choice

Finite-Shot Effects on Generalized Eigenvalue Stability

Scaling of QSE and Future Adaptive Strategies

Computational Methods

Molecular Geometries and Active Space Preparation

The dataset used comprised 28 closed-shell organic molecules from Thiel’s excited-state benchmark set [1]. The published cartesian nuclear coordinates were referenced directly and used without further optimisation.

For each molecule, Restricted Hartree-Fock calculations were performed using PySCF [2, 3] with the STO-3G basis set. This set of canonical molecular orbitals were optimised once and used consistently throughout all subsequent CASCI, VQE and QSE calculations in the study.

Active spaces (denoted $n_e e n_o o$) comprised of the $n_e/2$ occupied molecular spatial-orbitals closest to the HOMO and the $n_o - n_e/2$ virtual molecular spatial-orbitals closest to the LUMO. The remaining occupied orbitals were treated as frozen and doubly occupied, while the remaining virtual orbitals were excluded from calculations.

Exact Reference

Within each active space, the exact reference energies and wavefunctions were obtained from complete active-space configuration interaction (CASCI) calculations performed using PySCF. Calculations were separated by spin-projection sector M_S by setting the number of active α and β spin electrons to be

$$n_\alpha = \frac{n_e}{2} + M_S, n_\beta = \frac{n_e}{2} - M_S$$

Singlet states were obtained from the $M_S = 0$ sector, while triplet states were calculated in the $M_S = 0, +1, -1$ sectors. For each active-space spin projection sector, the determinant-space dimension, and therefore the maximum number of available roots, was $\binom{n_o}{n_\alpha} \binom{n_o}{n_\beta}$.

Where available, the lowest 11 singlet states (ground state + 10 excited states), and the lowest 10 triplet states (within each spin projection sector) were calculated. To ensure that the target number of states were obtained, three times the required number of CASCI roots were initially requested from the FCI solver, subject to the maximum dimension of the corresponding determinant space. The resulting roots were then filtered according to their $\langle \hat{S}^2 \rangle$ expectation values, retaining only roots that satisfy $|\langle \hat{S}^2 \rangle - S(S+1)| < 10^{-5}$ with $S = 0$ for singlets and $S = 1$ for triplets.

Throughout this study, wavefunctions were represented in a common active spin-orbital determinant basis by a state vector of length $\binom{2n_o}{n_e}$. This vector includes all determinants with n_e electrons distributed among the $2n_o$ active spin-orbitals (irrespective of their spin projections). Determinants were ordered lexicographically using an interleaved spin-orbital convention ($1\alpha, 1\beta, 2\alpha, 2\beta, \dots$).

PySCF represented each wavefunction as a two dimensional CI coefficient matrix with dimensions $\binom{n_o}{n_\alpha}$ by $\binom{n_o}{n_\beta}$, corresponding to the α and β electron occupation strings. These coefficients were embedded into the corresponding entries of the common state-vector representation. During this conversion, the fermionic phase arising from the permutation between PySCF's separated α and β electron ordering and the interleaved spin-orbital convention was included explicitly.

Quantum Computing Framework

All quantum-circuit calculations in this study were performed using classical simulations implemented with the Qibo and Qibochem [4, 5] frameworks. These frameworks were used to construct, represent and simulate the quantum circuits employed. A research fork of Qibochem was developed to support the computational workflow used in the study. The fork introduced configurable ansatz classes, optimised circuit evaluation routines, support for measuring spin observables, and dedicated functionality for constructing and sampling the projected matrices required for Quantum Subspace Expansion calculations.

All calculations in this work used ideal noiseless circuit simulations without modelling any hardware. Only finite-sampling noise arising from a limited number of measurement shots was considered in the relevant sections.

Hamiltonian and Spin Operators

For each active space, the electronic Hamiltonian was constructed in second quantisation as follows:

$$\hat{H} = E_{\text{nuc}} + \sum_{pq} h_{pq} \hat{a}_p^\dagger \hat{a}_q + \frac{1}{2} \sum_{pqrs} g_{pqrs} \hat{a}_p^\dagger \hat{a}_q^\dagger \hat{a}_s \hat{a}_r. \quad (11)$$

where E_{nuc} is the core nuclear repulsion energy and p, q, r, s index the active spin orbitals. h_{pq} and g_{pqrs} are the one and two-electron integral coefficients evaluated in the fixed canonical Hartree-Fock MO basis.

Spin operators were constructed in the same second quantisation basis as follows.

$$\hat{S}_z = \frac{1}{2} \sum_p (\hat{n}_{p\alpha} - \hat{n}_{p\beta}), \quad \hat{n}_{p\sigma} = \hat{a}_{p\sigma}^\dagger \hat{a}_{p\sigma}. \quad (12)$$

$$\hat{S}_+ = \sum_p \hat{a}_{p\alpha}^\dagger \hat{a}_{p\beta}, \quad \hat{S}_- = \sum_p \hat{a}_{p\beta}^\dagger \hat{a}_{p\alpha}. \quad (13)$$

$$\hat{S}^2 = \hat{S}_z^2 + \frac{1}{2} (\hat{S}_+ \hat{S}_- + \hat{S}_- \hat{S}_+). \quad (14)$$

All fermionic Hamiltonian, spin and excitation operators were mapped to qubit operators using the Jordan-Wigner (JW) transformation.

Ground State Reference

As introduced above, approximate ground-state reference wavefunctions were prepared using parameterised Unitary Coupled Cluster (UCC) Ansatz. In this study, 6 classes of Ansatz were used, namely: UCCSD, UCCSDSinglet, UCCGSD, and k -UpCCGSDSinglet with $k = 1, 2$, and 3. The Ansatz were characterised by the excitation generators included in the cluster operator $\hat{T}$, summarised in the following table.

Table 1: Classification of the 6 UCC ansatz families

Ansatz	Singles excitations	Doubles excitations	Spin- conserving	Generalised	Spin- adapted	Paired
UCCSD	✓	✓	✓			
UCCSDSinglet	✓	✓	✓		✓	
UCCGSD	✓	✓	✓	✓		
1UpCCGSDSinglet	✓	✓	✓	✓	✓	✓
2UpCCGSDSinglet	✓	✓	✓	✓	✓	✓
3UpCCGSDSinglet	✓	✓	✓	✓	✓	✓

For each ansatz, candidate singles and doubles transitions were first enumerated in fermionic form. Only canonical transitions that excite from a lower to high spin orbital were retained because the reverse transitions would already be included through its conjugate adjoint $\hat{T}^\dagger$ before exponentiation. All retained generators were spin conserving: single excitations preserved spin labels, while double excitations preserved the total number of α and β spin electrons.

The non-generalised UCCSD and UCCSDSinglet ansatz were restricted to occupied-to-virtual single and double excitations relative to the reference Hartree-Fock determinant [6]. In contrast, the generalised UCCGSD ansatz allowed generalised orbital transitions between occupied-occupied and virtual-virtual spin orbitals [7, 8]. To reduce the size of the generalised-double operator pool, mix-direction double excitations, in which one electron was promoted while the other was relaxed, were excluded. Across the *4e4o* to *8e8o* active spaces, this practical restriction reduced the number of independent parameters by 42.2–45.2% and the circuit depth by 46.4–47.6%.

The k -UpCCGSDSinglet ansatz combined generalised singles with generalised paired doubles. For each paired doubles, an $\alpha\beta$ electron pair was removed from one spatial orbital and created in another. The UCCSDSinglet and k -UpCCGSDSinglet constructions imposed spin-adapted parameter sharing between spin-orbital excitations that had the same spatial-orbital labels. For k -UpCCGSDSinglet, the complete generalised single and paired-double block was repeated k times, with an independent set of variational parameters assigned to each repetition.

Unless otherwise stated, all UCC operators were implemented using the first order Suzuki Trotter product approximation given in eq. (15).

$$e^{\hat{T} - \hat{T}^\dagger} \approx \prod_{\lambda} e^{\theta_{\lambda} (\hat{T}_{\lambda} - \hat{T}_{\lambda}^\dagger)}. \quad (15)$$

At every optimisation step, the circuit energy was evaluated as the noiseless statevector expectation value of the electronic Hamiltonian eq. (11). The variational parameters were optimised classically using the L-BFGS-B optimiser through Qibo, to minimise the ground state expectation value. For the k -UpCCGSDSinglet ansatz, the parameters associated with all k repeated layers were optimised simultaneously. Before optimisation, ansatz parameters were initialised based on the following hierarchy:

$$\text{MP2} \rightarrow \text{1-UpCCGSD} \rightarrow \text{2-UpCCGSD} \rightarrow \text{3-UpCCGSD} \rightarrow \text{UCCSDSinglet} \rightarrow \text{UCCSD} \rightarrow \text{UCCGSD}.$$

Parameters associated with excitation generators shared with the preceding ansatz were initialised with the previously optimised amplitudes, while newly introduced generators were initialised to zero. The initial occupied to virtual double-excitation parameters were estimated using second-order Moller-Plesset perturbation theory (MP2). For parameter transfer from a k -UpCCGSDSinglet ansatz to a subsequent ansatz, the optimized amplitudes associated with each generator were summed across the k layers and used as the initial amplitude for the corresponding generator.

Following optimisation, the resulting VQE parameters were stored as a fixed ground-state reference wavefunction. This fixed reference set was used consistently for all subsequent statevector and finite-shot QSE calculations.

QSE Operator Pools and Projected Matrices

As discussed in the introduction, singlet and triplet operator pools consisting of only one-body operators were used to separately build the singlet and triplet subspaces. The singlet operator pool $\mathcal{O}^S$ is defined as such:

$$\hat{O}_{ij}^S = \hat{a}_{i\alpha}^\dagger \hat{a}_{j\alpha} + \hat{a}_{i\beta}^\dagger \hat{a}_{j\beta}, \quad (16)$$

$$\mathcal{O}^S = \left\{ \hat{O}_{ij}^S \mid i, j \in \{1, \dots, n_o\} \right\}. \quad (17)$$

whereby, i and j to denote active spatial orbitals and the diagonal singlet operators with $i = j$ include number operators.

The triplet operator pool $\mathcal{O}^T$ is constructed using operators from all 3 spin projection sectors, and recombined into one pool as follows.

$$\hat{O}_{ij}^{T,0} = \frac{1}{\sqrt{2}} \left(\hat{a}_{i\alpha}^\dagger \hat{a}_{j\alpha} - \hat{a}_{i\beta}^\dagger \hat{a}_{j\beta} \right), \quad (18)$$

$$\hat{O}_{ij}^{T,+1} = \hat{a}_{i\alpha}^\dagger \hat{a}_{j\beta}, \quad \hat{O}_{ij}^{T,-1} = \hat{a}_{i\beta}^\dagger \hat{a}_{j\alpha}, \quad (19)$$

$$\mathcal{O}^T = \left\{ \hat{O}_{ij}^{T,m} \mid i, j \in \{1, \dots, n_o\}, m \in \{-1, 0, +1\} \right\}. \quad (20)$$

Each spatial-orbital-indexed operator in either pool was enumerated as a single element $\hat{O}_\mu$, such that μ and ν label arbitrary operator-pool elements rather than spatial orbitals. The bare QSE observables were defined as

$$\bar{s}_{\mu\nu} = \hat{O}_\mu^\dagger \hat{O}_\nu, \quad \bar{h}_{\mu\nu} = \hat{O}_\mu^\dagger \hat{H} \hat{O}_\nu, \quad \bar{z}_{\mu\nu} = \hat{O}_\mu^\dagger \hat{S}^2 \hat{O}_\nu. \quad (21)$$

In total, there are

These projected operator pools were used to build the upper triangular elements of the H and S matrix. The full matrices were then reconstructed by complex conjugation.

Finite-Shot Simulation

Grouping of commuting terms $\rightarrow$ greedy grouping

Multinomial sampling method with numpy (allow of $O(1)$ scaling)

Numerical Analysis and Evaluation Metrics

Generalised Eigenvalue Problem

Solving the problem

Define thresholding and dimension solve

stable dimensions (2e3o etc) -> Why need to filter even for SV

Change of basis to pvec form for projection calculations

Projection

Formula to calculate projection

Triplet Degeneracies

Grouping degenerate states Mean of projection Splitting

Non Index State Matching

hungarian algorithm

Conclusions

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Data Availability

Supporting Information

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